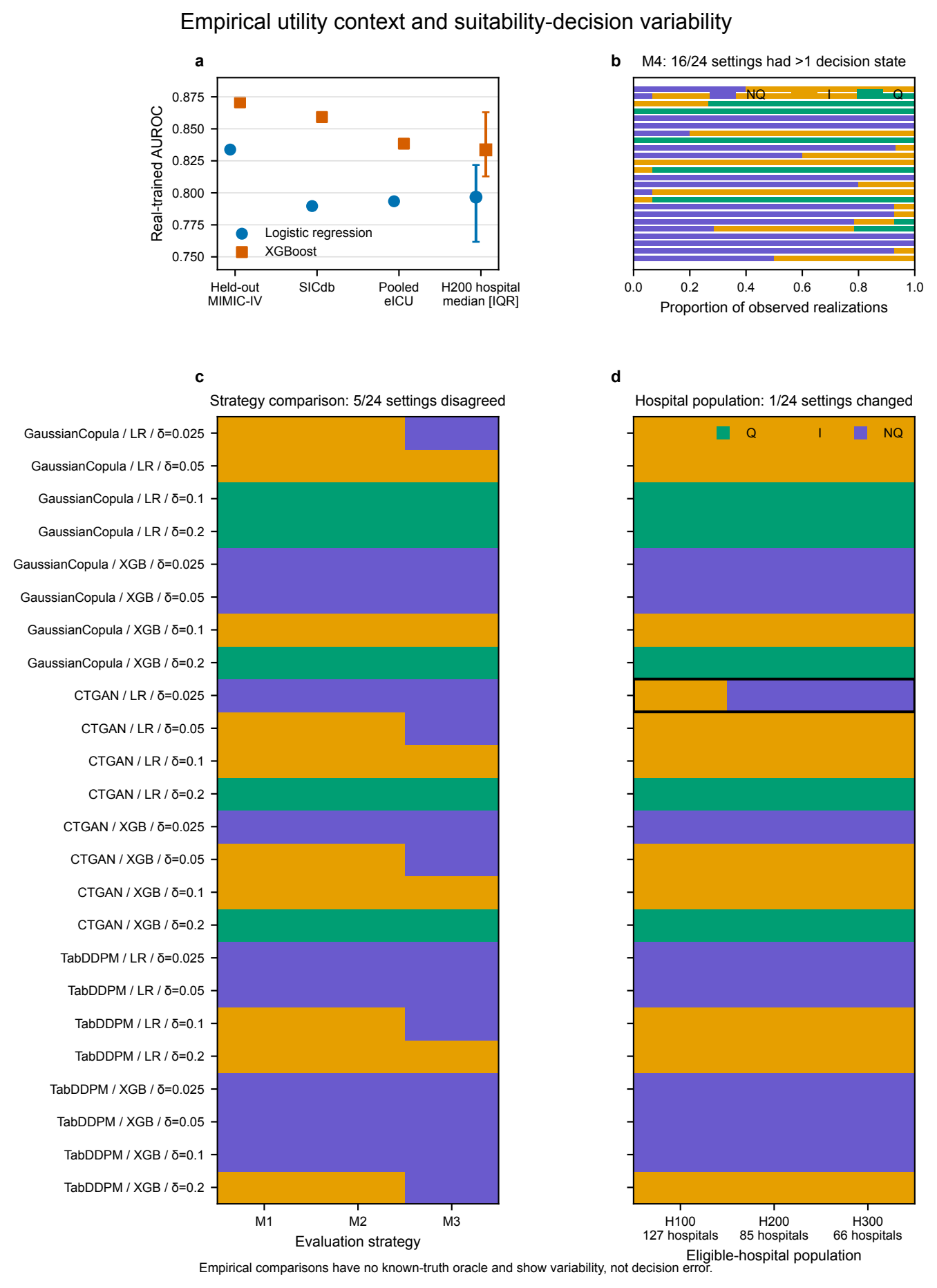
Empirical utility context and suitability-decision variability
a
Real-trained AUROC
0.875
0.850
0.825
0.800
0.775
0.750
Logistic regression
XGBoost
Held-out MIMIC-IV
SICdb
Pooled eICU
H200 hospital median [IQR]
b
M4: 16/24 settings had >1 decision state
NQ
I
Q
0.0
0.2
0.4
0.6
0.8
1.0
Proportion of observed realizations
c
Strategy comparison: 5/24 settings disagreed
GaussianCopula / LR / δ=0.025
GaussianCopula / LR / δ=0.05
GaussianCopula / LR / δ=0.1
GaussianCopula / LR / δ=0.2
GaussianCopula / XGB / δ=0.025
GaussianCopula / XGB / δ=0.05
GaussianCopula / XGB / δ=0.1
GaussianCopula / XGB / δ=0.2
CTGAN / LR / δ=0.025
CTGAN / LR / δ=0.05
CTGAN / LR / δ=0.1
CTGAN / LR / δ=0.2
CTGAN / XGB / δ=0.025
CTGAN / XGB / δ=0.05
CTGAN / XGB / δ=0.1
CTGAN / XGB / δ=0.2
TabDDPM / LR / δ=0.025
TabDDPM / LR / δ=0.05
TabDDPM / LR / δ=0.1
TabDDPM / LR / δ=0.2
TabDDPM / XGB / δ=0.025
TabDDPM / XGB / δ=0.05
TabDDPM / XGB / δ=0.1
TabDDPM / XGB / δ=0.2
M1
M2
M3
Evaluation strategy
d
Hospital population: 1/24 settings changed
Q
I
NQ
H100
127 hospitals
H200
85 hospitals
H300
66 hospitals
Eligible-hospital population
Empirical comparisons have no known-truth oracle and show variability, not decision error.